

IEDA 4331 GROUP PROJECTS

1 Objective

Utilize the pricing techniques from IEDA 4331 and relevant data sources to design a financial derivative tailored for targeted investors (customers) for selected underlying assets. Provide detailed pricing and risk management strategies.

2 Data source

- Bond/equity/cryptocurrencies data: <https://finance.yahoo.com>
- Various financial data: Bloomberg at HKUST Library
- Cryptocurrency data: www.blockchain.com; <https://www.delta.exchange>; <https://www.deribit.com>
- Other sources

3 Example topics

- Asia option for BTC/ETH
- Barrier option for BTC/ETH
- Vanilla option for BTC/ETH
- Other derivatives for other underlying assets

4 Requirements on project report

1. Introduction: Provide a summary of the research objective, methodology, and findings.
2. Methodology and Data: Describe your quantitative analysis method, data overview, and estimation techniques utilized in the project.
3. Project Findings: Present data analysis, pricing results, quantitative analysis, and discussions.
4. Additional statement: If your project involves the assistance of AI or ChatGPT, please clearly outline in the appendix how you utilized the AI or ChatGPT.

5 Group Formation

- Each group can have a maximum of three members.
- The group with only one member will be subjected to a 5% penalty.

6 Timeline

- Time for reporting group formation: March 12, 2026. Submit the names of group members to the TA's email (wwubx@connect.ust.hk).
- Project report deadline: May 20, 2025.